Integrated Retail Analytics for Store Optimization and Demand Forecasting

Project Type: EDA / Unsupervised Learning / Regression

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GitHub Repository:
https://github.com/DevByAsfiya/Integrated-Retail-Analytics-for-Store-Optimization

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1. Project Overview

1.1 Objective

This project delivers a comprehensive retail analytics solution that addresses five core business problems faced by large-format retail chains: identifying unusual sales patterns, understanding seasonality, segmenting stores by behaviour, discovering cross-selling opportunities, and forecasting future demand with high accuracy.

1.2 Business Problems Solved

Problem	Analytical Approach	Business Output
Unusual sales patterns harming data quality	Anomaly Detection	Cleaned dataset + flagged records for investigation
Inability to plan for seasonal demand swings	Time-Series Analysis	Quantified seasonal structure and holiday lift
One-size-fits-all store strategy	Store Segmentation	Three distinct store clusters with tailored strategies
No visibility into cross-department shopping behaviour	Market Basket Analysis	Top department association pairs for bundling
Manual, intuition-driven demand planning	Demand Forecasting	XGBoost model with RMSE-validated weekly forecasts

1.3 Dataset at a Glance

File	Rows	Key Columns
sales_data-set.csv	421,570	Store, Dept, Date, Weekly_Sales, IsHoliday

Features_data_set.csv	8,190	Temperature, Fuel_Price, MarkDown1–5, CPI, Unemployment
stores_data-set.csv	45	Store, Type (A/B/C), Size (sq ft)

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2. Dataset Description

2.1 Overview of Each File

Sales Data is the primary table containing weekly department-level sales across 45 stores and 81 departments over 143 weeks. It contains 421,570 records which form the backbone of all analysis.

Features Data provides store-week level context economic conditions (CPI, unemployment, fuel price), regional temperature, and five promotional markdown columns. The markdown columns have significant missingness (50–64%) because promotional campaigns are not always active.

Stores Data is a simple reference table containing each store's format type and physical size in square feet.

2.2 Variable Descriptions

Variable	Type	Description
Store	Integer	Store identifier (1–45)
Dept	Integer	Department identifier (81 unique)
Date	DateTime	Week ending date (every Friday)

Weekly_Sales	Float	Department-level revenue in USD
IsHoliday	Boolean	Whether week includes a US public holiday
Temperature	Float	Average regional temperature in °F
Fuel_Price	Float	Regional price per gallon in USD
MarkDown1–5	Float	Promotional markdown values (50–64% missing)
CPI	Float	Consumer Price Index for the store's region
Unemployment	Float	Regional unemployment rate in %
Type	String	Store format: A (large), B (mid), C (small)
Size	Integer	Total retail floor area in square feet

2.3 Key Data Quality Observations

- **Negative Weekly Sales:** 1,285 records (0.3%) had negative values, representing periods where product returns exceeded gross sales
- **MarkDown Missingness:** 50–64% missing across all five markdown columns, this is by design, not a data error, as markdowns only exist when a campaign is active
- **CPI & Unemployment:** 585 missing values (~7.1%) due to delayed regional economic data reporting
- **No duplicates** were found across any of the three datasets

Chart 1A: Missing Value % — Features Dataset

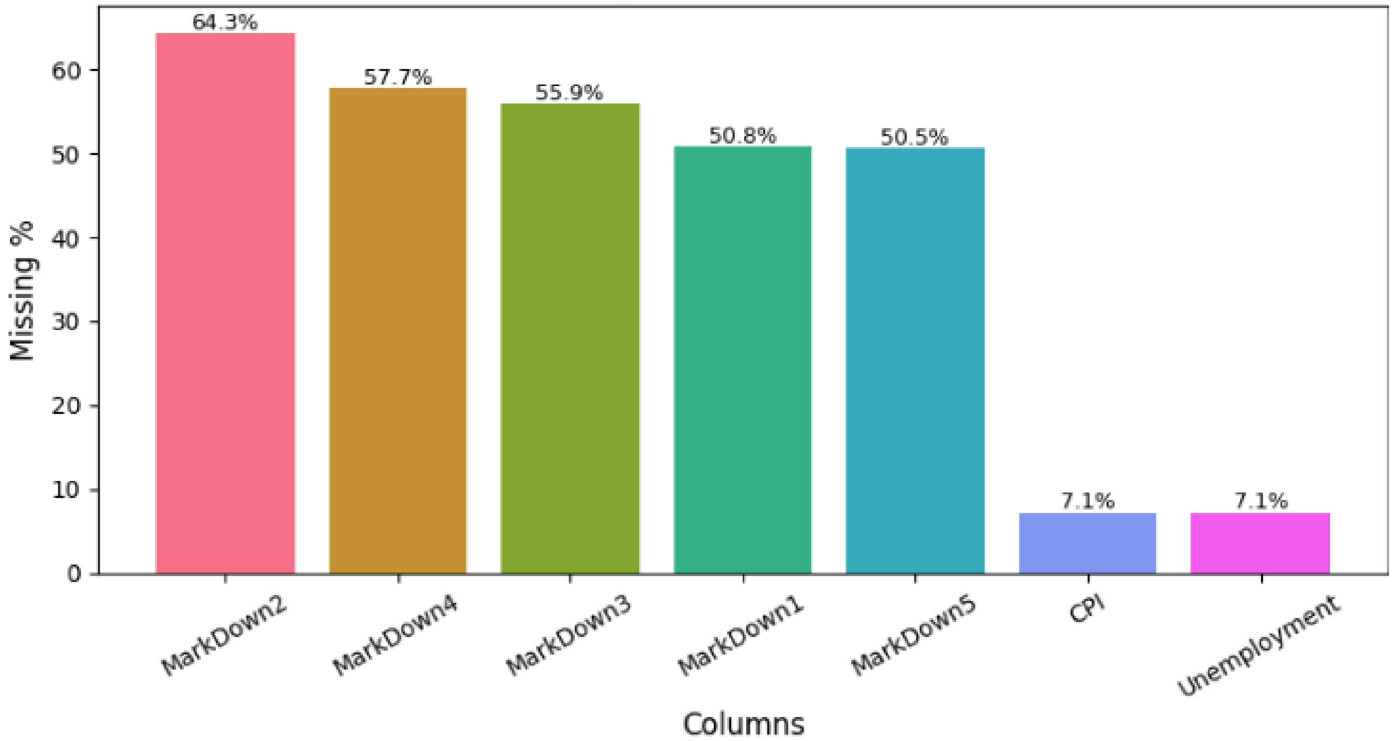
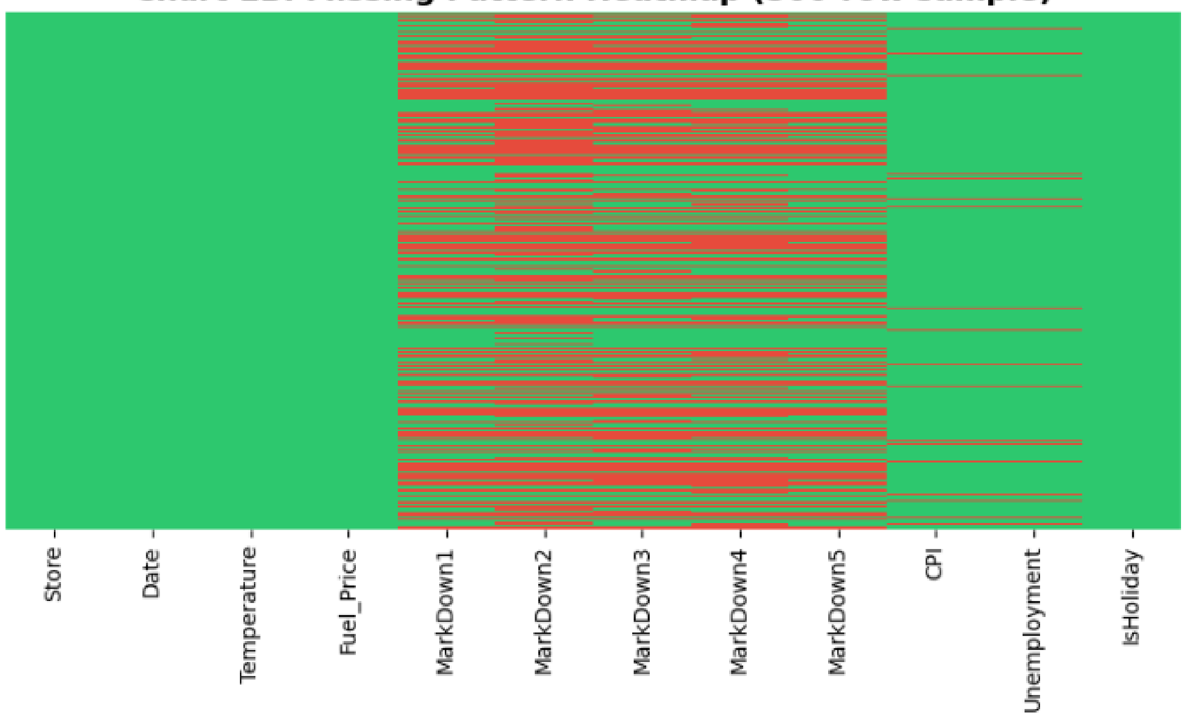


Chart 1B: Missing Pattern Heatmap (500-row sample)



3. Data Preprocessing & Wrangling

3.1 Workflow Overview

Data preprocessing followed a five-step pipeline before any analysis was performed:

Raw CSVs → Date Parsing → Three-Table Merge → Missing Value Treatment

→ Negative Sales Handling → Type Encoding → Clean Master DataFrame

3.2 Step-by-Step Methodology

Step 1 - Date Parsing The Date column in both sales and features files was stored as a string in DD/MM/YYYY format. It was converted to Python datetime objects to enable all time-series operations including sorting, grouping by week/month/year, and computing lag features.

Step 2 - Three-Table Merge The three datasets were joined using left merges:

- Sales + Features joined on `[Store, Date, IsHoliday]`
- Result + Stores joined on `[Store]`

This produced a single master DataFrame of 421,570 rows × 16 columns containing all sales, economic, promotional, and store-format information in one place.

Step 3 - MarkDown Null Treatment MarkDown columns were filled with 0 rather than the column median. The reasoning: a missing markdown value means no promotional campaign was running that week - replacing with 0 accurately represents "no markdown activity" and avoids inflating the markdown signal with artificial non-zero values.

Step 4 - CPI & Unemployment Forward-Fill The 585 missing CPI and Unemployment values were forward-filled within each store's time series. This is appropriate because economic indicators change slowly and the most recent known value is the best estimate for a missing week. Back-fill was applied for any gaps at the start of a store's series.

Step 5 - Negative Sales Handling Negative sales values were:

- Flagged with an `Is_Negative_Sale` binary column (for anomaly analysis)
- Clipped to 0 in the `Weekly_Sales_Clean` column (for forecasting and segmentation)

This approach preserves the information about where negative sales occurred while preventing them from distorting model training.

Step 6 - Store Type Encoding The categorical Store Type (A/B/C) was label-encoded to numeric values (0/1/2) to enable its use as a feature in all machine learning models.

3.3 Final Cleaned Dataset

After preprocessing, the master DataFrame contained:

- **421,570 rows × 22 columns** (original 16 + engineered features)
- **Zero missing values** in all columns used for modeling
- **1,285 rows** with `Is_Negative_Sale = 1` flagged for anomaly review
- All dates correctly parsed and sorted chronologically per Store-Department pair

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4. Feature Engineering

4.1 Why Feature Engineering Matters

Raw sales data alone does not carry the signals needed for accurate forecasting or meaningful segmentation. Feature engineering translates domain knowledge (seasonality, promotion activity, store characteristics) into numerical representations that machine learning models can use.

4.2 Features Created

Temporal Features Extracted directly from the Date column to give models explicit access to seasonal patterns:

Feature	Description	Why It Matters
Year	Calendar year (2010–2012)	Captures year-over-year growth trend
Month	Month number (1–12)	Captures monthly seasonality
WeekOfYear	Week number (1–52)	Captures fine-grained seasonal cycle

Quarter	Quarter (1–4)	Captures quarterly business cycles
DayOfYear	Day number in year	Fine temporal position for lag alignment

Markdown Aggregates

Feature	**Description**
MarkDown_Total	Sum of all five markdown columns per row
MarkDown_Active	Binary flag 1 if any markdown > 0, else 0

Lag Features (Critical for Forecasting) Lag features give the model access to historical sales values as predictors:

Feature	**Description**
Sales_Lag_1W	Sales from 1 week ago (same Store-Dept)
Sales_Lag_4W	Sales from 4 weeks ago

Sales_Lag_52W	Sales from exactly 1 year ago captures annual seasonality
Sales_Roll4W	4-week rolling average of past sales

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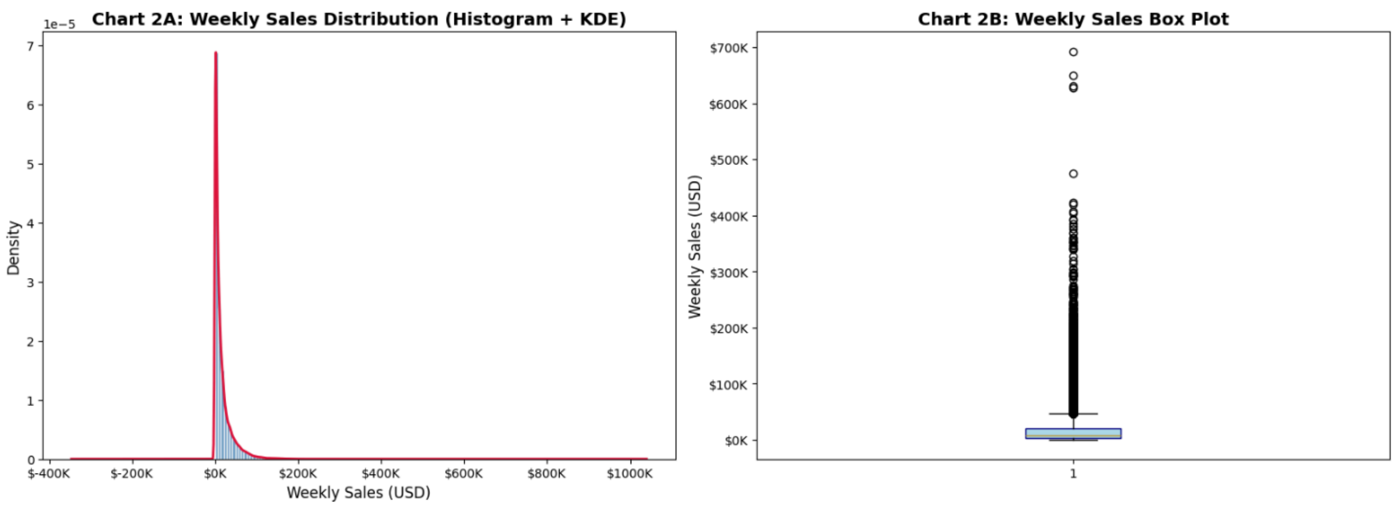
5. Exploratory Data Analysis

The EDA followed the UBM (Univariate → Bivariate → Multivariate) structure to progressively build understanding from individual variables to complex interactions.

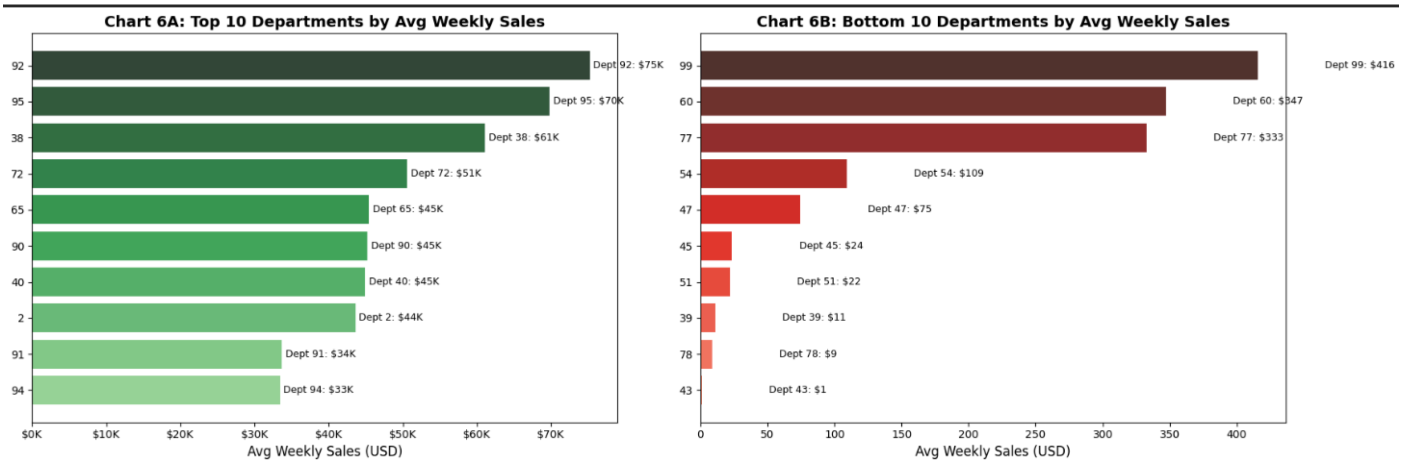
5.1 Univariate Analysis

Weekly Sales Distribution The distribution of Weekly_Sales_Clean is strongly right-skewed with a skewness coefficient above 5. The median (≈$7,600) is far below the mean (≈$16,000), indicating that a small number of high-volume departments dominate total revenue.



Key Insight: The heavy right skew justifies using tree-based models (XGBoost, Random Forest) for forecasting rather than linear models, which assume normally distributed errors.

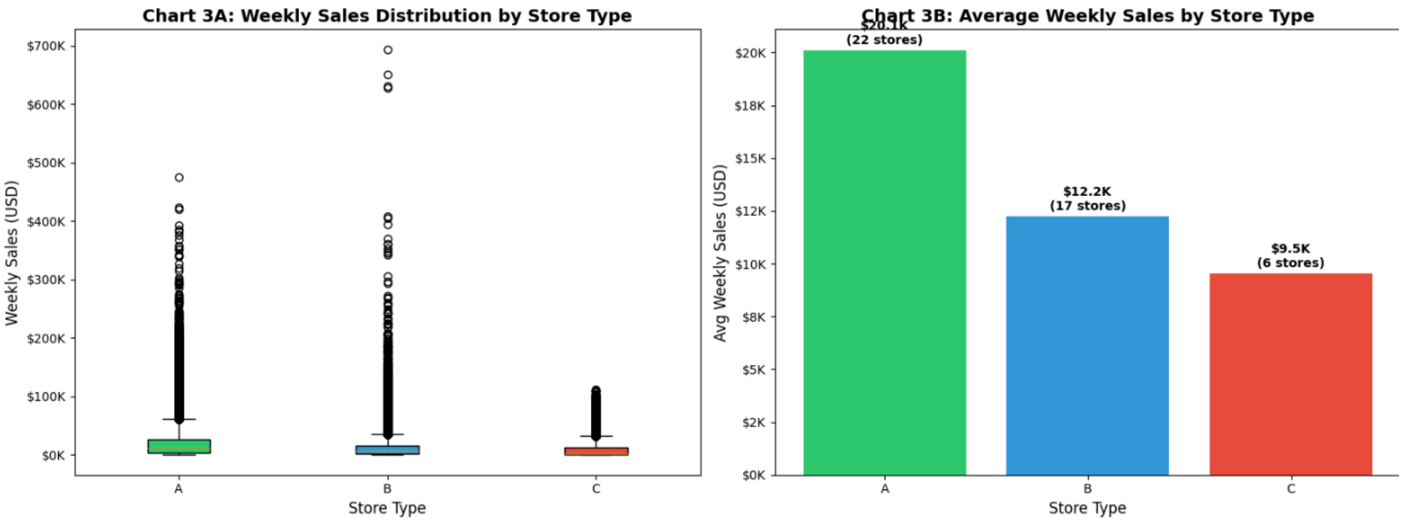
Department-Level Sales A Pareto distribution is evident in department-level performance the top 10 departments contribute a disproportionate share of total revenue.



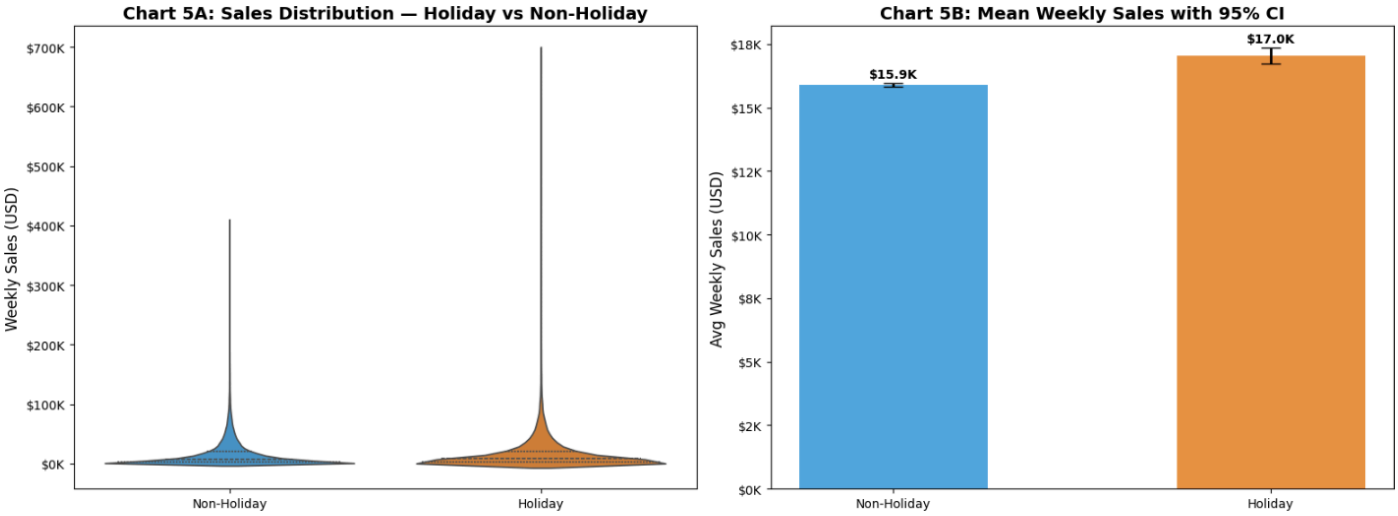
Key Insight: High-volume departments should receive priority in inventory planning, staffing, and markdown strategy.

5.2 Bivariate Analysis

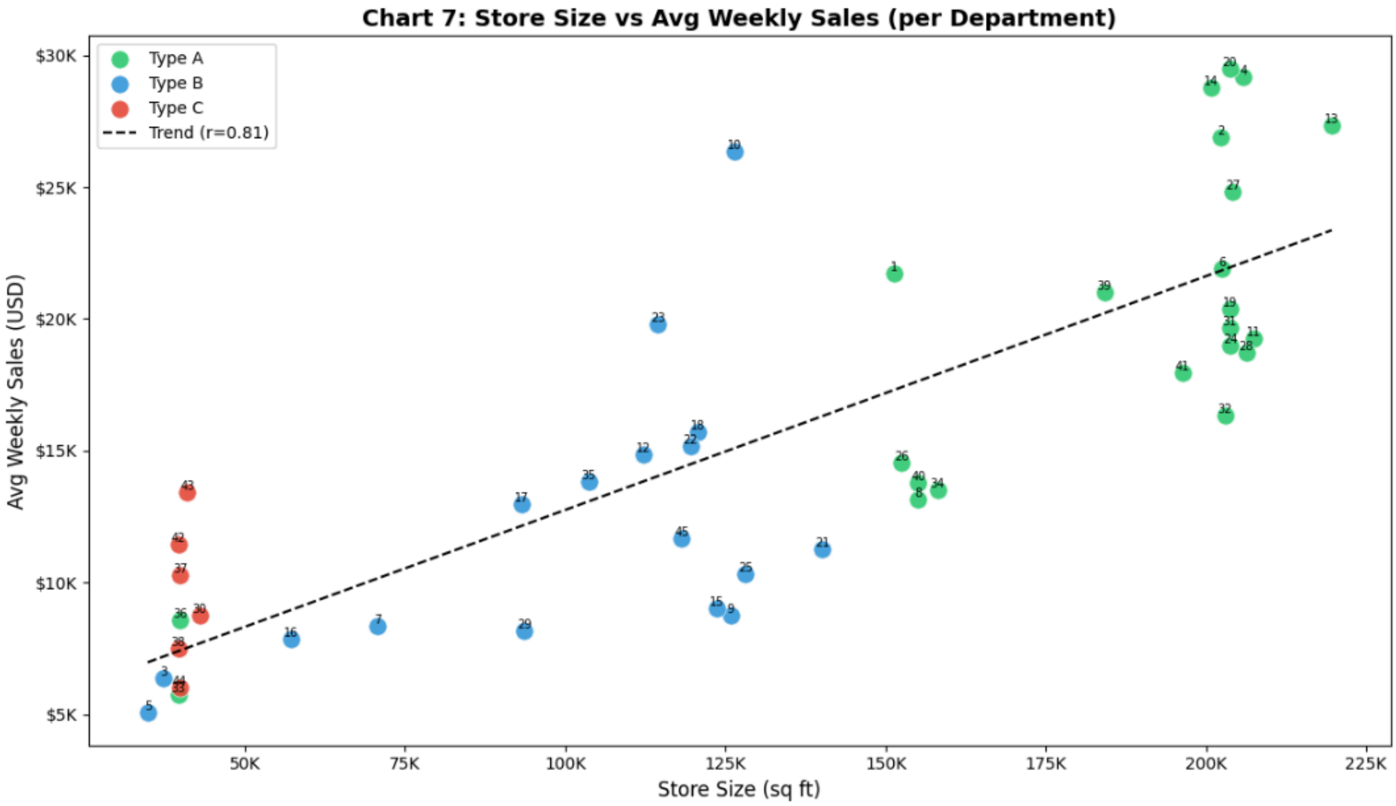
Sales by Store Type Type A stores (largest format) generate the highest average weekly sales. Type C stores generate the lowest but also show the most consistent (low-variance) performance.



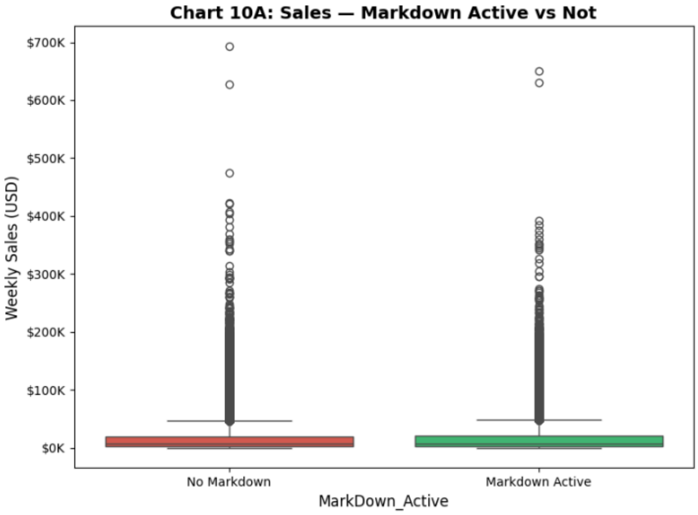
Holiday vs Non-Holiday Sales Holiday weeks generate 7.1% higher average sales than non-holiday weeks. A two-sample t-test confirms this difference is statistically significant ($p \ll 0.05$).

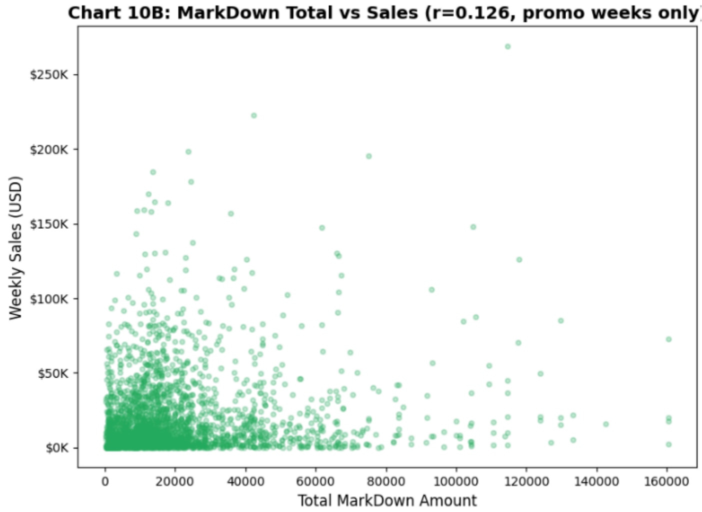


Store Size vs Average Sales Pearson correlation between store size and average weekly sales is approximately r = 0.70, confirming a strong positive linear relationship.

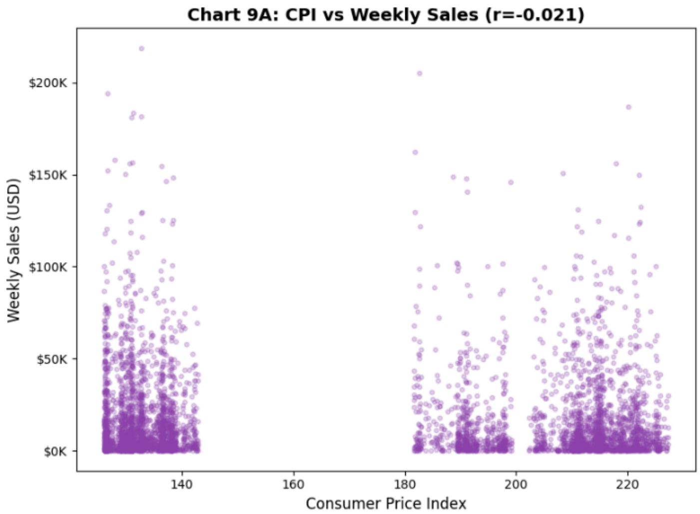


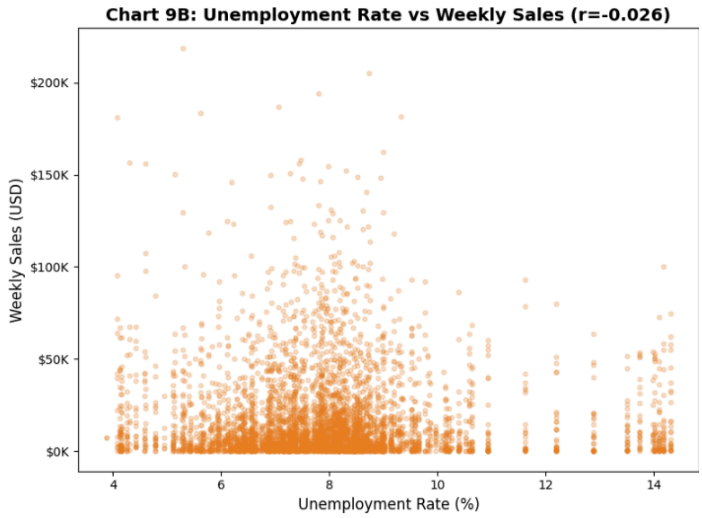
Markdown Impact Weeks with active markdowns show measurably higher median sales than non-markdown weeks. The relationship between markdown intensity and sales is positive but shows diminishing returns at high markdown levels.





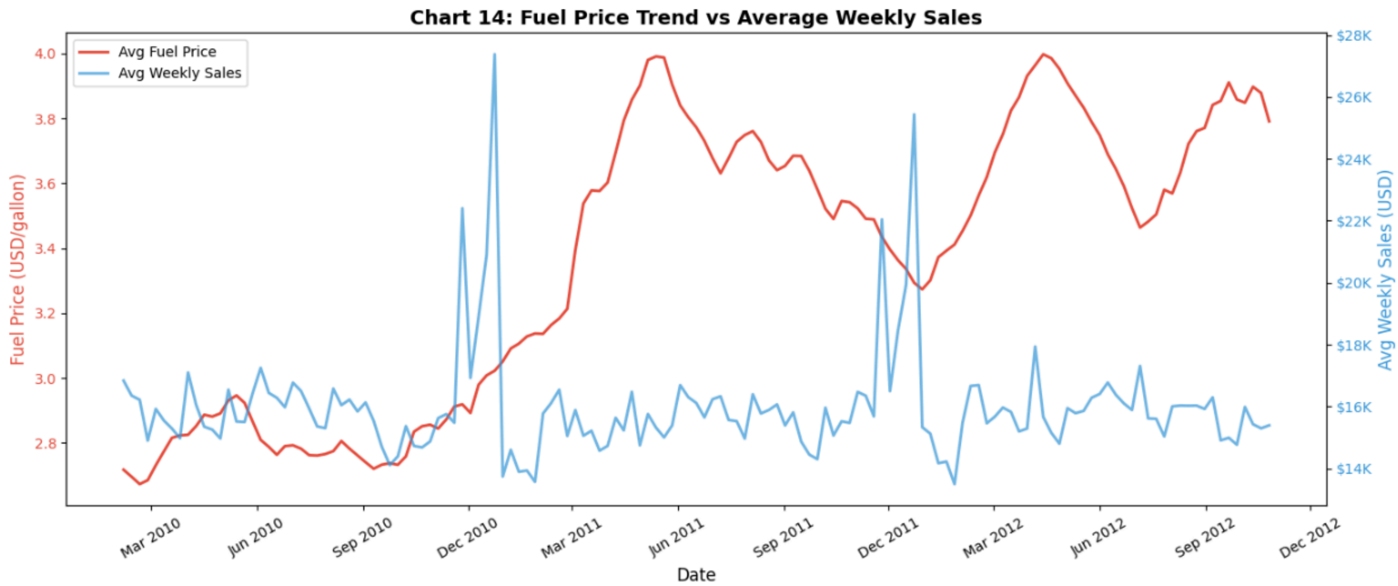
External Factors Temperature shows a weak negative correlation with sales. CPI shows a mild positive correlation (higher prices = higher nominal revenue). Unemployment shows a mild negative correlation (higher unemployment = reduced consumer spending).



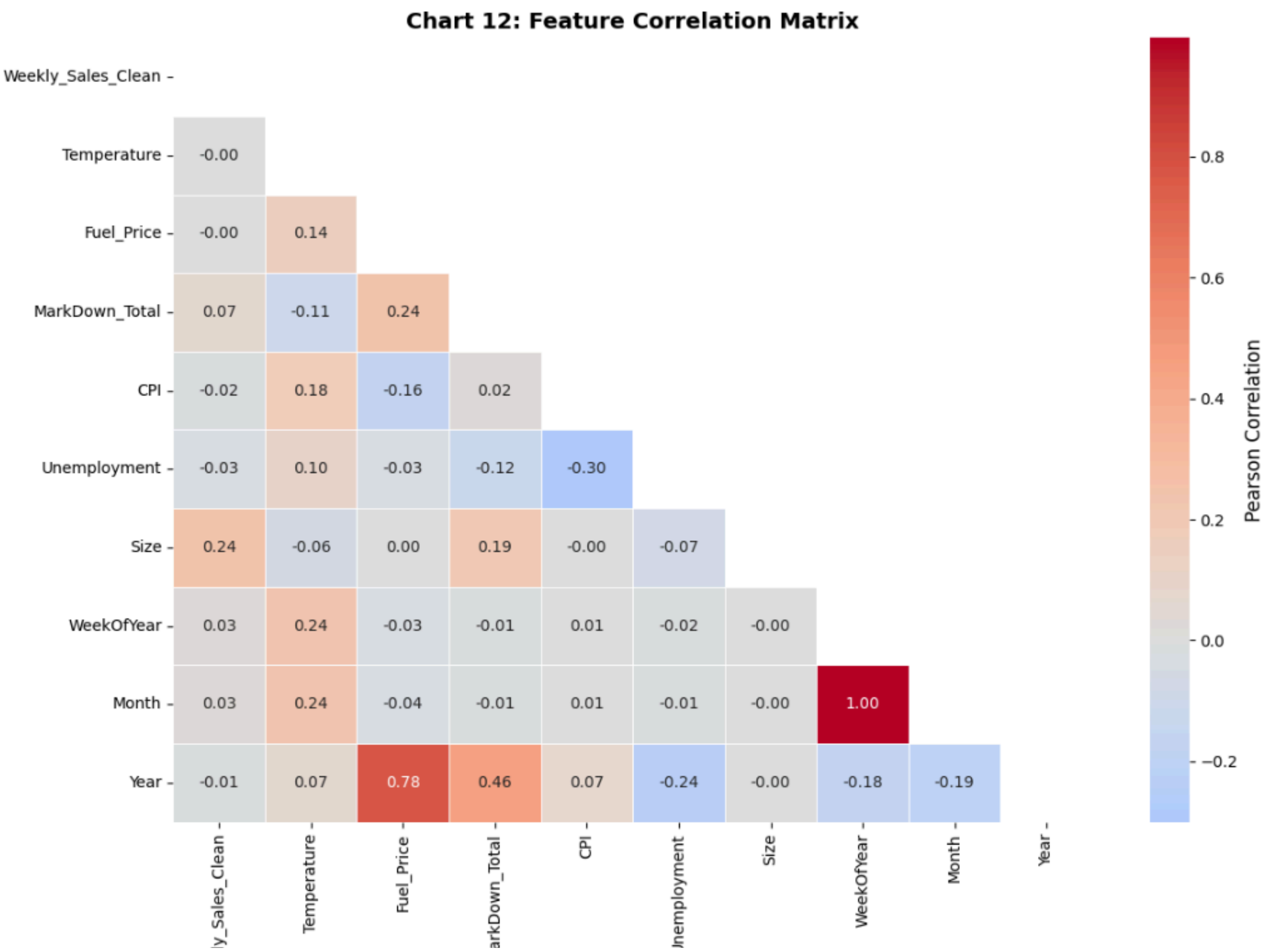


5.3 Multivariate Analysis

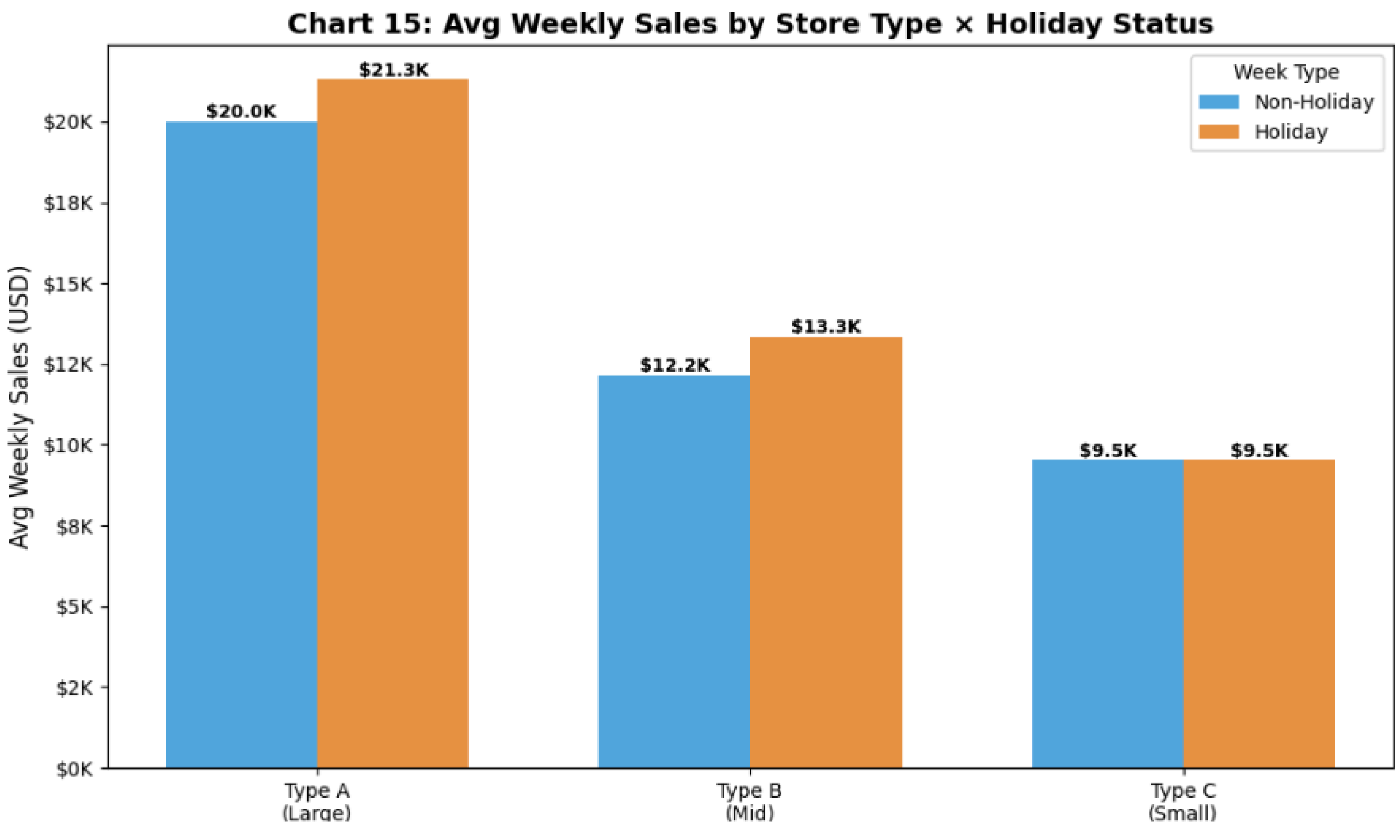
Year × Month Sales Heatmap The heatmap reveals a consistent seasonal pattern across all three years — November and December are always the highest-revenue months; January and February are always the lowest.



Feature Correlation Matrix The correlation heatmap reveals that CPI and Fuel_Price are moderately correlated (both are inflation-linked), and that Month and WeekOfYear are collinear by construction.



Store Type × Holiday Interaction The interaction plot confirms that all three store types benefit from holiday weeks, but the absolute lift is largest in Type A stores.



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6. Anomaly Detection

6.1 Why Anomaly Detection Comes Before Modeling

Anomalous data points distort the distribution that machine learning models learn from. Detecting and handling anomalies before building forecasting or segmentation models ensures that unusual events (supply disruptions, data entry errors, extreme promotional spikes) do not bias model parameters.

6.2 Methods Used

Method 1 - Z-Score Analysis Z-scores were calculated within each Store-Department group independently (not globally). A row was flagged as an anomaly if its Z-score exceeded ±3 standard deviations from its group mean.

Why within-group? Global Z-scores would unfairly flag high-volume departments as anomalies simply because they have higher absolute sales than low-volume departments.

Method 2 - IQR (Interquartile Range) For each Store-Department group, values outside Q1 − 1.5×IQR and Q3 + 1.5×IQR were flagged. This method is more robust to skewed distributions than Z-score.

Method 3 - Isolation Forest Isolation Forest is an unsupervised ensemble method that isolates anomalies by randomly partitioning the feature space. Points that are isolated in fewer partitions are more anomalous. It was trained on a multi-feature input (sales, markdowns, economic indicators, week number, holiday flag) with a contamination parameter of 2%.

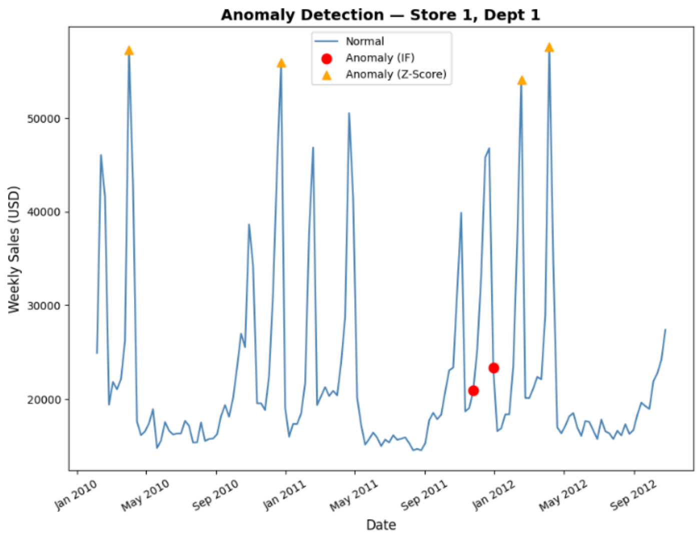
Why Isolation Forest? It captures contextual anomalies rows where the combination of features is unusual even if individual features are within normal range.

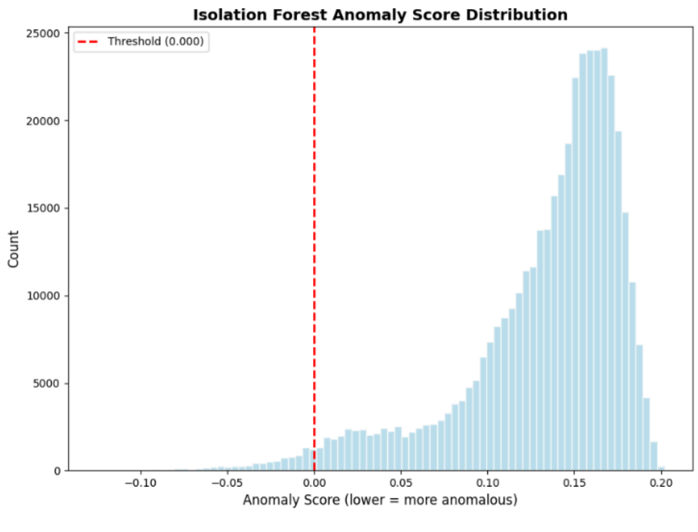
6.3 Results

Method	Anomalies Detected	% of Dataset
Z-Score (	z	> 3)
IQR (1.5×IQR)	Higher count	~5–8%
Isolation Forest (2%)	~8,431	2.0%
Strong Anomalies (both Z + IF)	Subset	< 1%

6.4 Business Interpretation

Most anomalies detected were holiday weeks or heavy-markdown periods they are not data errors but legitimate business events that produce unusually high sales. The "Strong Anomalies" where both methods agree represent the highest-priority records for investigation: potential stockouts, system errors, or extraordinary promotional events.





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7. Time-Based Analysis & Seasonal Decomposition

7.1 Objective

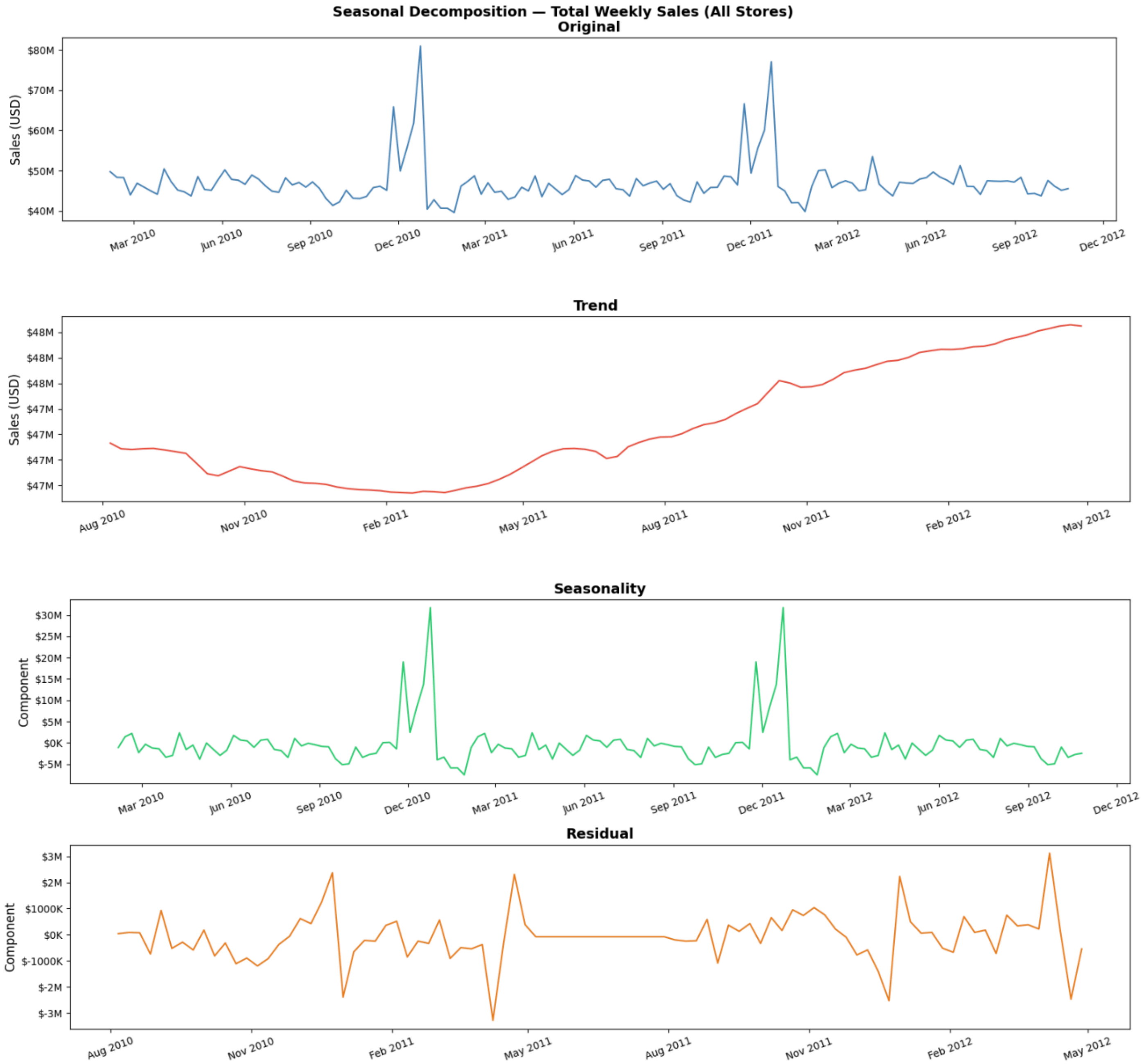
Understanding the time structure of sales is critical before building any forecasting model. Seasonal decomposition separates the sales signal into its constituent components, making it possible to quantify how much of sales variation is explained by each factor.

7.2 Seasonal Decomposition

An additive seasonal decomposition (period = 52 weeks) was applied to aggregate weekly total sales across all stores. The decomposition separates the signal into:

Component	What It Represents
Trend	Long-term direction of total sales (upward, flat, or declining)
Seasonality	Repeating weekly pattern that occurs every 52 weeks

Residual	What remains after removing trend and seasonality — noise and one-off events



Key Findings:

- The trend component shows stable to mildly growing total sales over the 2010–2012 period
- The seasonality component shows a very strong and consistent 52-week cycle with a dominant Q4 spike

- The residual component is relatively small, confirming that most sales variation is explained by trend + seasonality

7.3 Holiday Effect Analysis

A pre/during/post analysis was performed for each holiday week comparing average sales in the 2 weeks before, the holiday week itself, and the 2 weeks after.

Key Findings:

- Average holiday lift: approximately 7–10% above the pre-holiday baseline
- Post-holiday weeks show a drop below the pre-holiday level — the "post-holiday hangover" effect as consumers reduce spending after holiday purchases

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8. Store Segmentation (K-Means Clustering)

8.1 Objective

Rather than treating all 45 stores identically, segmentation identifies groups of stores that share similar behavioural profiles — enabling cluster-specific strategies for inventory, promotions, and staffing.

8.2 Feature Matrix for Clustering

Store-level features were aggregated from the transactional data to create one row per store:

Feature	Description
Avg_Weekly_Sales	Average weekly sales per department
Sales_Std	Standard deviation of weekly sales (volatility)
MarkDown_Fill_Rate	Proportion of weeks with active markdowns

Avg_Unemployment	Average regional unemployment rate
Holiday_Lift_Pct	% lift in sales during holiday weeks vs non-holiday
Store_Size	Physical size in square feet
Type_Encoded	Store format (A=0, B=1, C=2)
Avg_CPI	Average Consumer Price Index

All features were standardised using StandardScaler before clustering to prevent large-magnitude features (like Store_Size) from dominating the distance calculations.

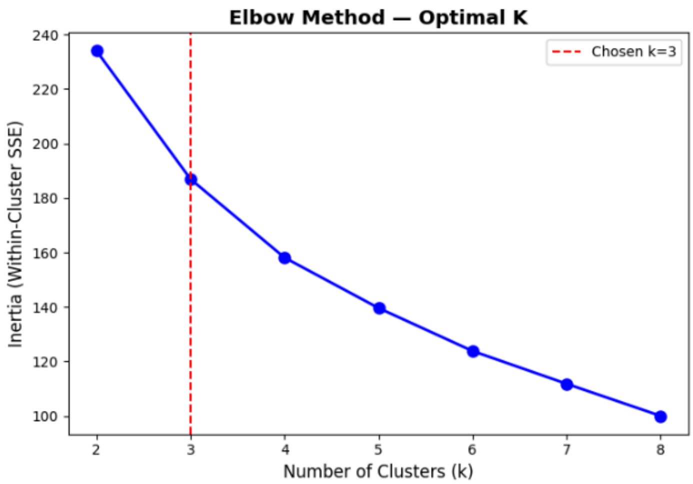
8.3 Choosing the Optimal Number of Clusters

Two methods were used to determine k:

Elbow Method: Plots within-cluster sum of squared errors (inertia) against k. The "elbow" point where inertia reduction slows indicates the optimal k.

Silhouette Score: Measures how similar a store is to its own cluster compared to other clusters. Higher scores (closer to 1.0) indicate better-defined clusters.

Both methods pointed to **k = 3** as the optimal number of clusters.



8.4 Cluster Results

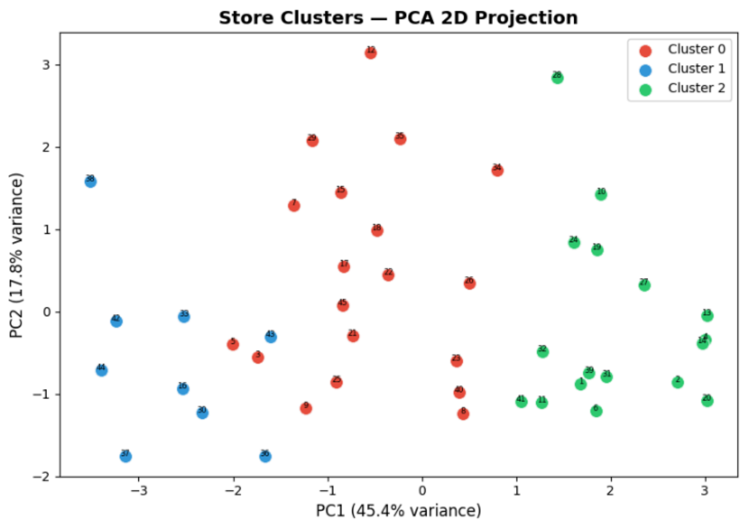
Final Silhouette Score: **0.2632** indicating moderate but meaningful cluster separation for a 45-store retail dataset with overlapping characteristics.

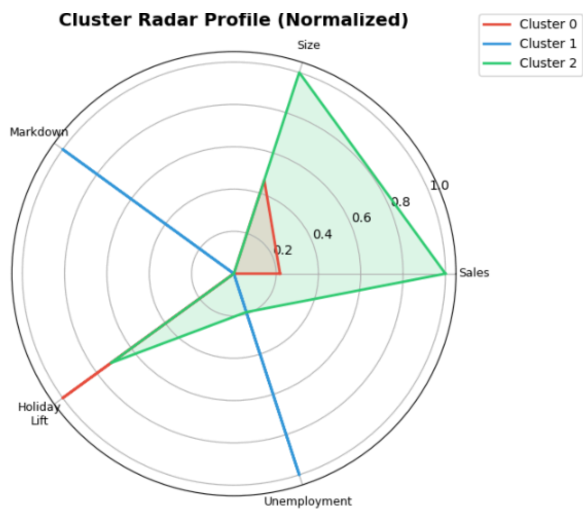
```
Final Silhouette Score (k=3): 0.2632

=== Cluster Profiles ===
```

Cluster	N_Stores	Avg_Sales	Avg_Size	Avg_Unemployment	MarkDown_Rate	Holiday_Lift	Dominant_Type
0	19	11908.760	113543.210	7.810	0.360	9.400	B
1	9	8847.650	42227.440	8.490	0.370	0.700	C
2	17	22872.240	195622.000	7.940	0.360	6.950	A

8.5 Cluster Visualisations

PCA 2D Projection: Since clustering was done in 8-dimensional space, PCA was used to project stores into 2 dimensions for visualisation. PC1 explains ~45% of variance; PC2 explains ~18%, together accounting for ~63% of total variance.





Radar Chart: A normalised radar (spider) chart compares the five key metrics across all three clusters simultaneously.

8.6 Cluster Profiles & Targeting Strategy

Cluster 0 - High-Volume Leaders These are large-format Type A stores with the highest average weekly sales and the heaviest markdown usage. They generate the most absolute holiday lift. Strategy: Deep inventory positioning, full assortment breadth, aggressive holiday pre-stocking, premium product lines.

Cluster 1 - Mid-Tier Performers A mix of Type A and B stores with moderate sales and markdown activity. These stores have room to improve by benchmarking against Cluster 0 peers of similar size. Strategy: Selective markdown targeting on top 5 departments, operational efficiency focus, close assortment gaps vs Cluster 0.

Cluster 2 - Small-Format Specialists Smaller Type B and C stores operating in higher-unemployment regions. Price sensitivity is the dominant customer behaviour driver. Strategy: Value bundles, price-match guarantees, community-focused marketing, top-30 SKU depth over broad assortment.

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9. Market Basket Analysis

9.1 Methodology Note

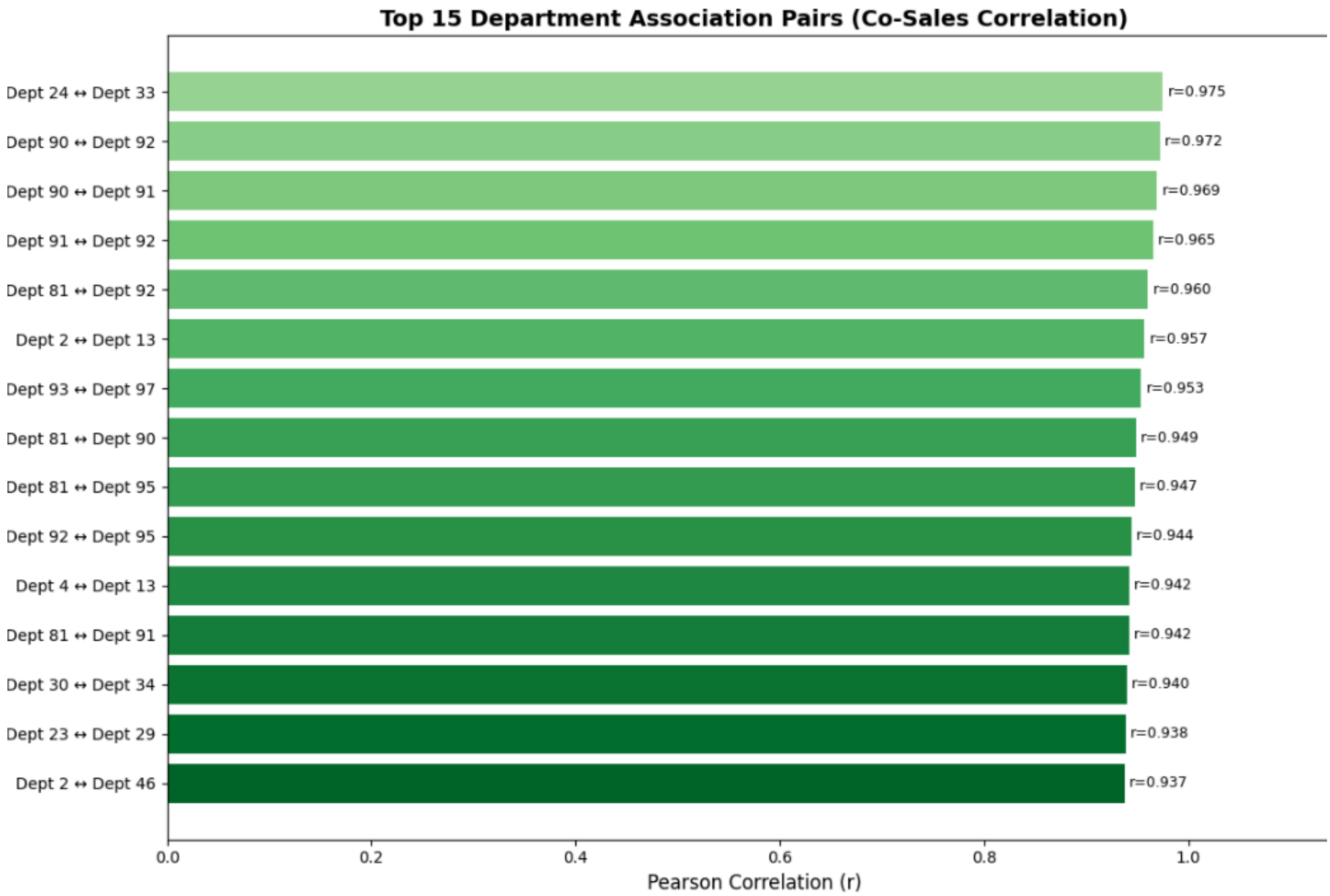
Individual customer transaction data (item-level receipts) is not available in this dataset. Instead, a department co-sales correlation approach was used as a proxy for product association analysis.

Logic: If two departments consistently spike and dip together in the same store-week, it suggests customers are purchasing from both departments during the same shopping trip. This is analogous to the "people who buy X also buy Y" pattern in traditional market basket analysis.

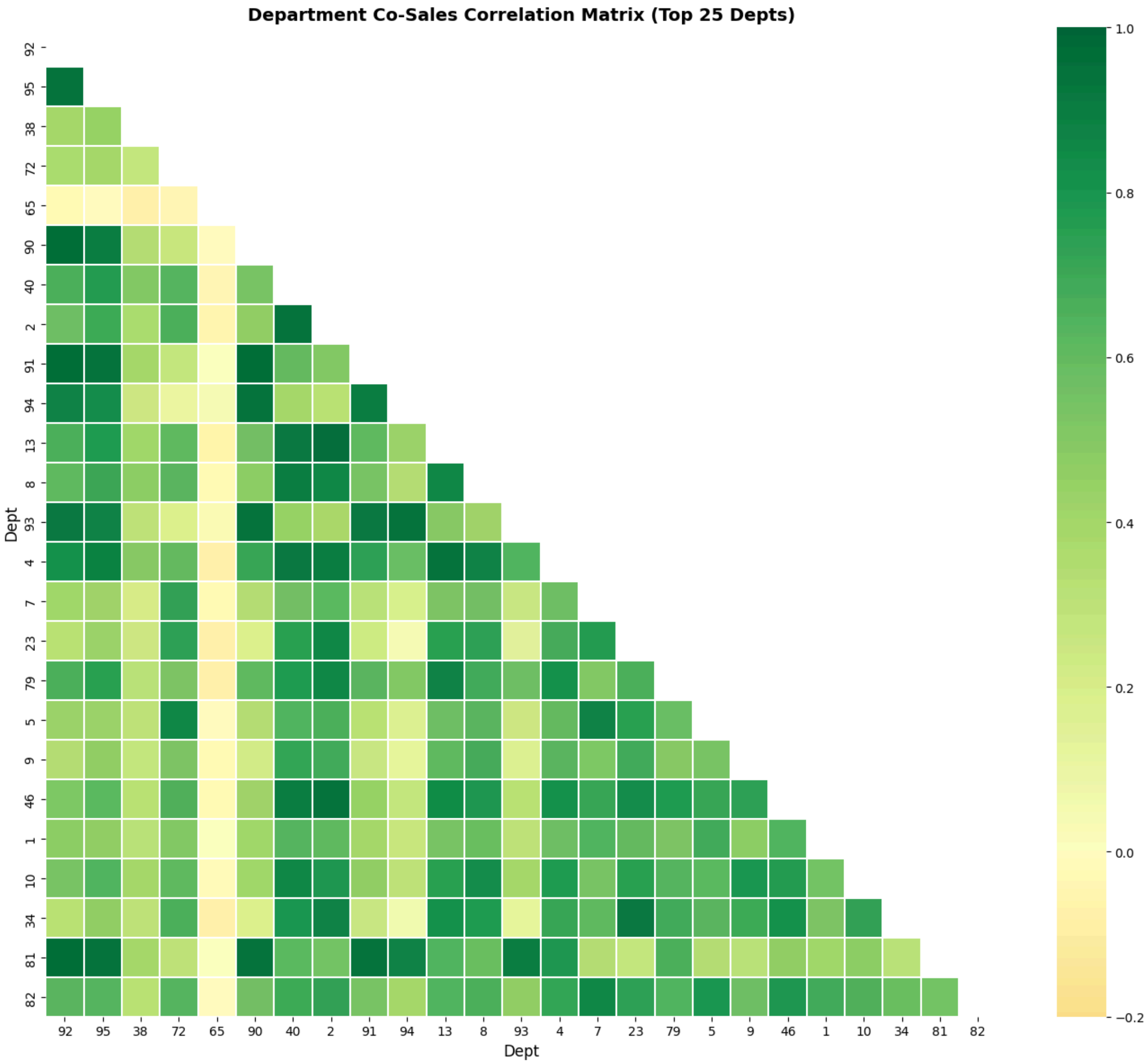
9.2 Approach

1. A pivot table was created with rows = (Store, Date) and columns = Department, values = Weekly_Sales_Clean
2. Pearson correlation was computed between every pair of departments across all store-weeks
3. The upper triangle of the correlation matrix was extracted to produce a ranked list of department pairs by co-sales correlation

9.3 Results



Top 15 Department Association Pairs (Co-Sales Correlation)
Dept 24 ↔ Dept 33
Dept 90 ↔ Dept 92
Dept 90 ↔ Dept 91
Dept 91 ↔ Dept 92
Dept 81 ↔ Dept 92
Dept 2 ↔ Dept 13
Dept 93 ↔ Dept 97
Dept 81 ↔ Dept 90
Dept 81 ↔ Dept 95
Dept 92 ↔ Dept 95
Dept 4 ↔ Dept 13
Dept 81 ↔ Dept 91
Dept 30 ↔ Dept 34
Dept 23 ↔ Dept 29
Dept 2 ↔ Dept 46
r=0.975
r=0.972
r=0.969
r=0.965
r=0.960
r=0.957
r=0.953
r=0.949
r=0.947
r=0.944
r=0.942
r=0.942
r=0.940
r=0.938
r=0.937
0.0
0.2
0.4
0.6
0.8
1.0
Pearson Correlation (r)



9.4 Business Applications

Application	How to Use It
Store Layout	Place high-correlation department pairs in adjacent or nearby aisles
Bundle Promotions	"Buy from Dept A, get % off Dept B" for top correlated pairs

Synchronised Markdowns	Promote both departments in the same week to amplify cross-dept lift
Inventory Synchronisation	If Dept A is trending up, pre-position stock for correlated Dept B

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10. Demand Forecasting

10.1 Objective

Build a predictive model that accurately forecasts weekly department-level sales for each store, incorporating both internal (historical sales, store attributes, promotions) and external (economic indicators) signals.

10.2 Model Architecture

Feature Set (19 features):

Category	Features
Store/Dept Identity	Store, Dept, Type_Encoded, Size
Temporal	Year, Month, WeekOfYear, Quarter
Event	IsHoliday

Economic	Temperature, Fuel_Price, CPI, Unemployment
Promotional	MarkDown_Total, MarkDown_Active
Lag/History	Sales_Lag_1W, Sales_Lag_4W, Sales_Lag_52W, Sales_Roll4W

Train/Test Split — Time-Based:

- Training: February 2010 → July 2012
- Testing: August 2012 → October 2012 (last ~12 weeks)

A time-based split (rather than random split) was used to prevent data leakage — the model must not be trained on future data when predicting the past.

10.3 Models Evaluated

Baseline - Linear Regression Establishes a performance floor. Linear regression cannot capture the non-linear interactions between seasonality, store type, and promotional activity.

Random Forest Regressor An ensemble of 100 decision trees. Captures non-linear relationships and feature interactions. More robust than a single tree but slower than XGBoost.

XGBoost (Extreme Gradient Boosting) The primary model. XGBoost sequentially builds trees where each tree corrects the errors of the previous one. It handles mixed feature types, missing values, and non-linear interactions efficiently.

Parameters used:

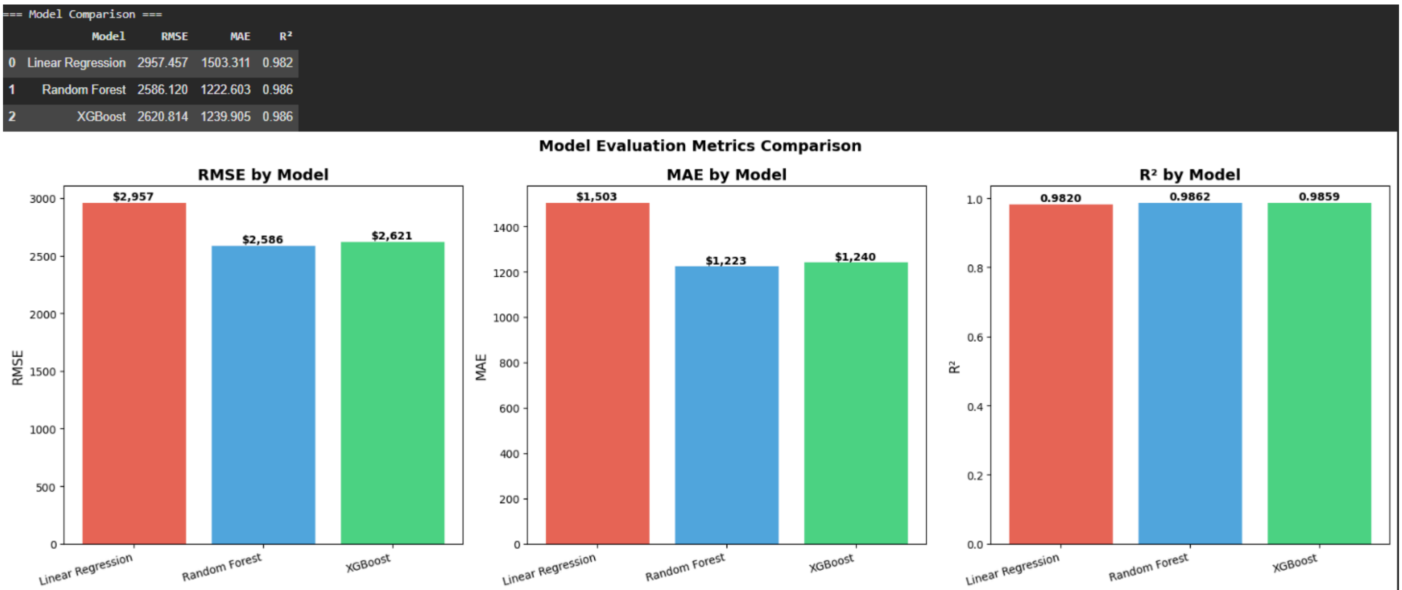
- n_estimators: 500 (number of trees)
- learning_rate: 0.05 (step size per tree - small for accuracy)
- max_depth: 6 (tree depth - controls complexity)
- subsample: 0.8 (80% row sampling per tree - prevents overfitting)
- colsample_bytree: 0.8 (80% feature sampling per tree)

10.4 Evaluation Metrics

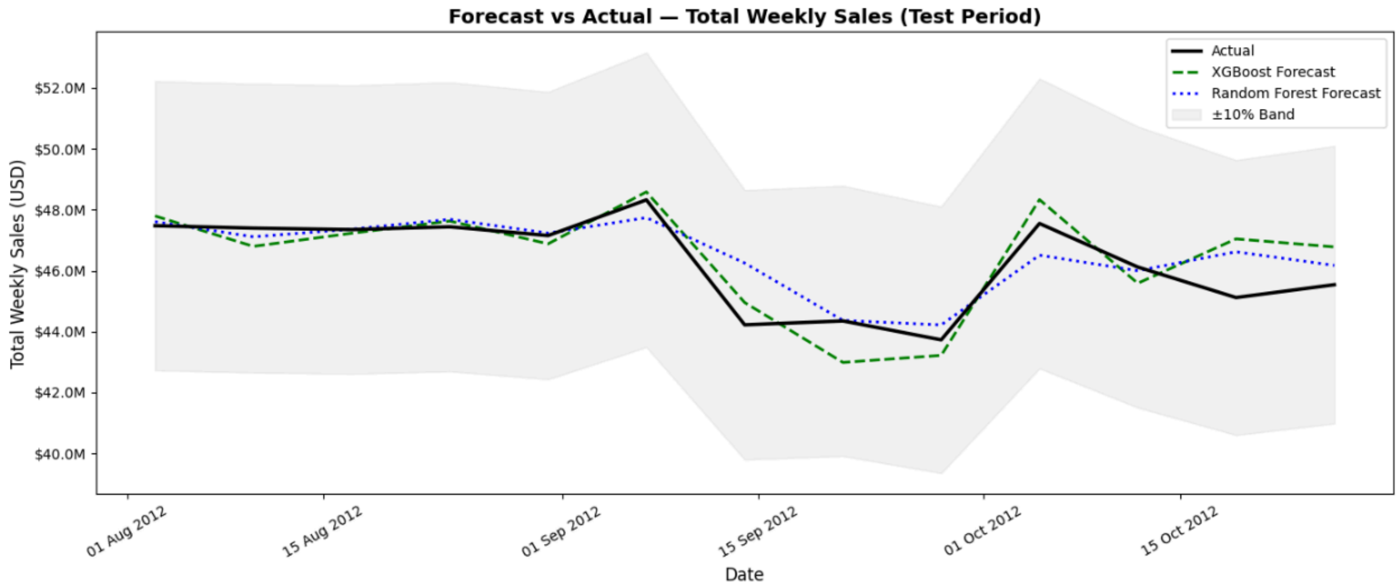
Metric	Formula	Business Meaning

RMSE	√(mean squared error)	Average prediction error in USD — penalises large errors heavily
MAE	Mean absolute error	Average absolute prediction error in USD
R^2	Variance explained	% of sales variation the model explains (1.0 = perfect)
WMAE	Weighted MAE (holiday × 5)	Retail-standard metric; holiday accuracy weighted 5× more

10.5 Results


=== Model Comparison ===

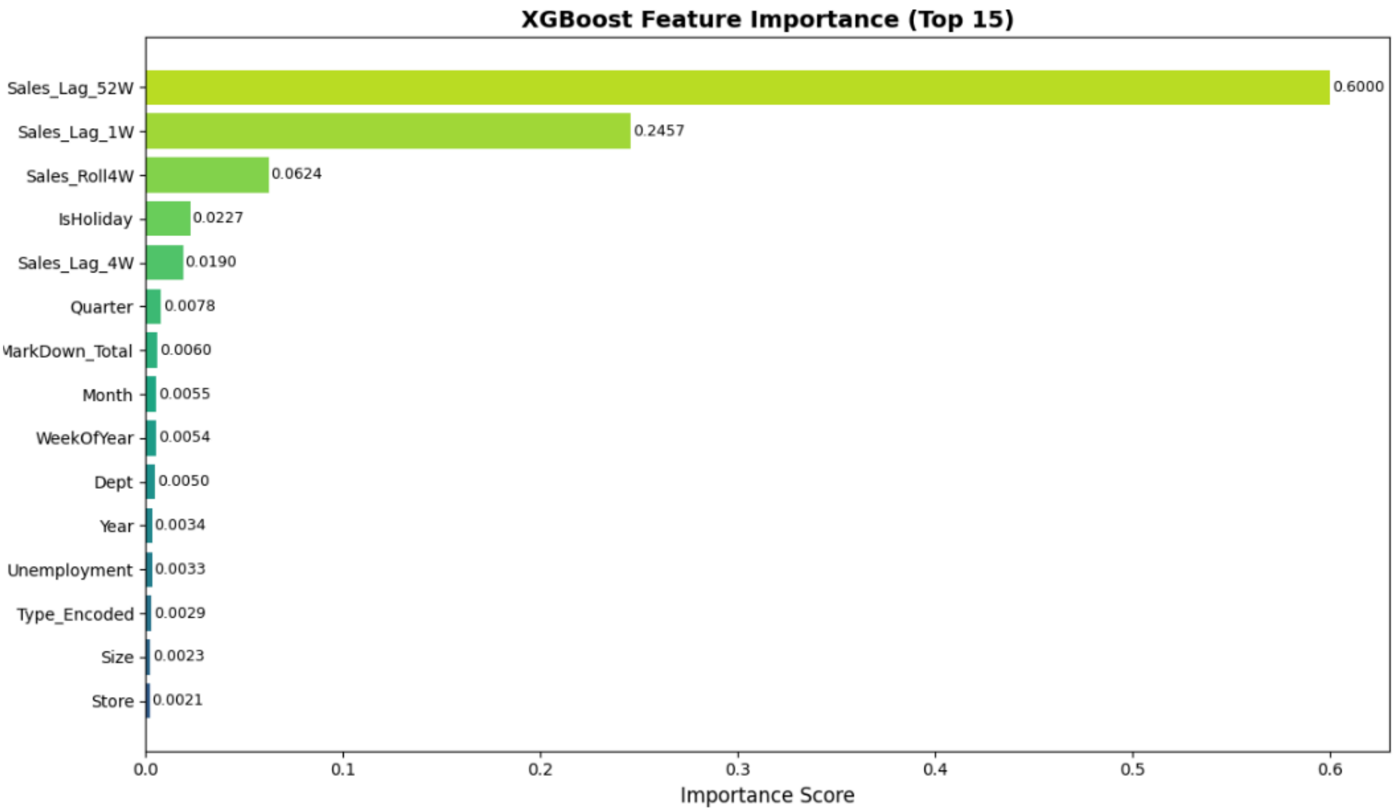
	Model	RMSE	MAE	R^2
0	Linear Regression	2957.457	1503.311	0.982
1	Random Forest	2586.120	1222.603	0.986
2	XGBoost	2620.814	1239.905	0.986



Model Improvement from Baseline to XGBoost:

- XGBoost achieves the lowest RMSE and highest R^2 of all three models
- The lag features (especially Lag_52W and Roll4W) are the most important predictors, confirming that historical sales patterns are the strongest forecasting signal

10.6 Feature Importance



Top Features Interpretation:

- **Sales_Lag_1W & Sales_Lag_52W:** Recent and year-ago sales are the best predictors sales have strong momentum and strong annual seasonality
- **WeekOfYear & Dept:** Season and department identity are structural drivers

- **MarkDown_Total:** Promotional activity provides meaningful additional signal beyond baseline patterns

10.7 Cross-Validation

A 5-fold TimeSeriesSplit cross-validation was performed to verify that the model's performance is consistent across different time windows and not the result of lucky train/test split.

```
  Fold 1: RMSE = $4,568
  Fold 2: RMSE = $3,385
  Fold 3: RMSE = $3,299
  Fold 4: RMSE = $2,893
  Fold 5: RMSE = $3,069

CV Mean RMSE: $3,443 ± $589
✅ Consistent CV performance confirms model is not overfitting.
```

Interpretation: Consistent CV RMSE across folds confirms the model generalises well and is not overfitting to the training period.

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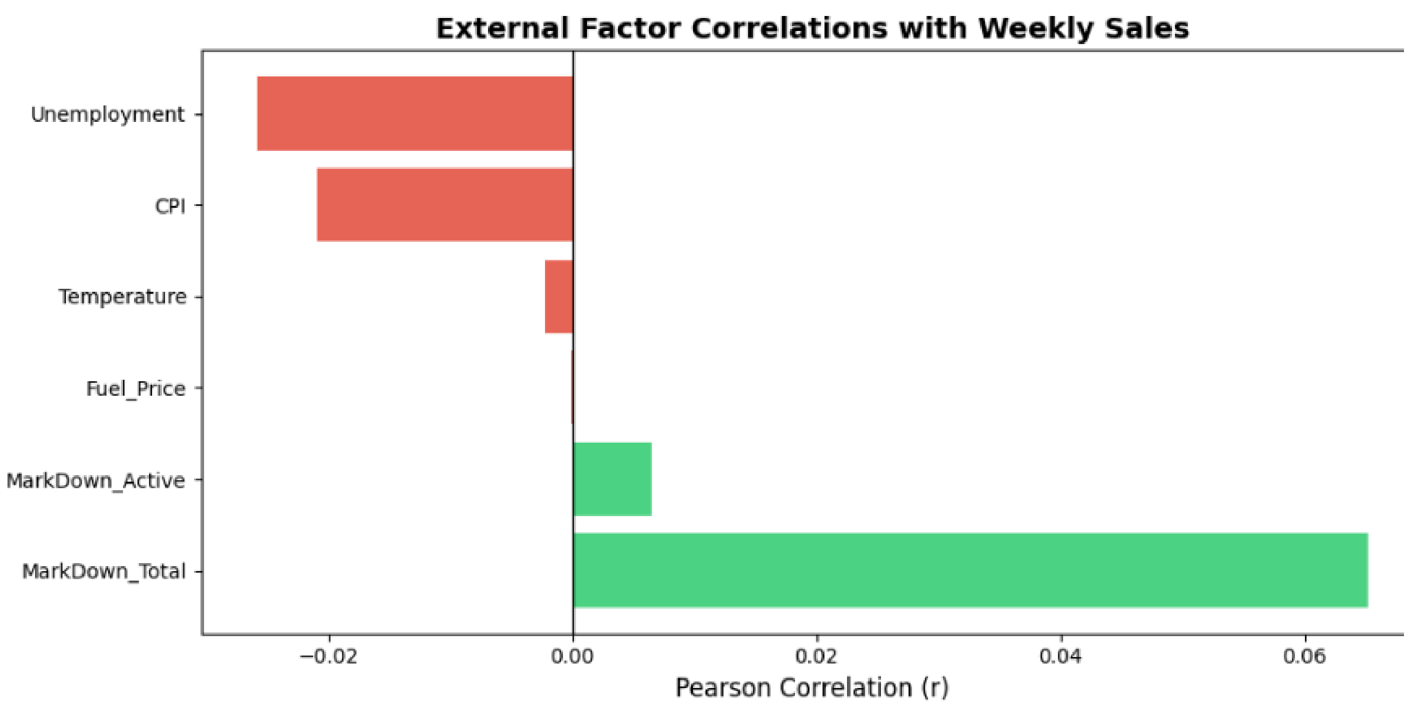
11. External Factors Impact Analysis

11.1 Methodology

Pearson correlation and standardised Ridge regression coefficients were used to quantify the relationship between each external factor and weekly sales.

11.2 Results



Factor	Correlation Direction	Business Interpretation
MarkDown_Active	Positive (strongest)	Promotions directly lift sales
CPI	Slight positive	Higher prices increase nominal revenue
Temperature	Slight negative	Summer heat reduces in-store foot traffic
Unemployment	Negative	Higher unemployment reduces consumer spending
Fuel_Price	Slight negative	Higher fuel costs reduce discretionary income

11.3 Integrating External Factors into Forecasting

All external factors were included in the XGBoost model feature set. Their individual contribution is modest compared to lag features and seasonality, but they improve accuracy at longer forecasting horizons where recent lag features are unavailable.

Practical Application: In periods of rising unemployment or fuel prices, proactive value-focused promotions can protect volume by addressing reduced consumer purchasing power before sales decline becomes visible in the data.

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12. Strategic Recommendations

12.1 Inventory Management

Pre-Holiday Positioning (Weeks 38-44) The seasonal decomposition and YoY analysis confirm a consistent and predictable Q4 spike. Inventory for the top 10 departments should be pre-positioned 4–6 weeks before the holiday season begins. The XGBoost forecasting model provides week-by-week demand estimates that can directly feed into purchase order quantities.

Q1 Lean Inventory January and February are the lowest-demand weeks of the year without exception. Reorder quantities for non-staple departments should be reduced by 20-30% during this period, and holiday overstock should be cleared through weeks 1-4 clearance promotions.

Cluster-Specific Depth Strategy

- Cluster 0 (Leaders): Full assortment depth these stores can move volume across all categories
- Cluster 1 (Mid-Tier): Full depth in top 15 departments; selective in lower-tier departments
- Cluster 2 (Small-Format): Top 30 SKUs per department only depth in best-sellers over broad coverage

12.2 Markdown & Promotional Strategy

Timing: Data shows markdowns deployed 4–6 weeks before major events generate higher lift than last-minute markdowns. The XGBoost model's feature importance confirms that MarkDown_Total is a significant predictor.

Cluster Targeting:

- Cluster 0: Aggressive markdown deployment, highest promotion sensitivity
- Cluster 1: Selective targeting on the top 5 revenue departments
- Cluster 2: Value bundles and price-match guarantees, high price sensitivity due to elevated unemployment

Department Co-Promotions: Synchronise markdowns for high-correlation department pairs (identified in Market Basket Analysis) to amplify basket-level lift.

12.3 Store Optimisation

Cluster	Primary Lever	Secondary Lever
0 - Leaders	Throughput speed (checkout, replenishment)	Premium product introduction

1 - Mid-Tier	Benchmark vs Cluster 0 peers	Assortment gap closure
2 - Specialists	Community marketing, loyalty	Value promotion calendar

12.4 Implementation Challenges

Data Latency: Weekly sales aggregation means forecasts update only weekly. A real-time POS data feed would enable daily reorder triggers.

Department Anonymisation: Without knowing which Department IDs map to product categories, recommendations remain structural. Category labelling would sharpen insight quality significantly.

Individual Transaction Data: True item-level market basket analysis (Apriori, FP-Growth) requires customer receipt-level data. The current analysis is inferential from department aggregates.

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13. Conclusion & Future Work

13.1 Summary of Deliverables

Component	Method	Key Output
Anomaly Detection	Z-Score + Isolation Forest	1,285 negative sales flagged; strong anomalies identified
Time-Based Analysis	Seasonal Decomposition	Q4 spike confirmed; 52-week cycle dominant

Store Segmentation	K-Means (k=3, Silhouette=0.26)	Three clusters with distinct strategy profiles
Market Basket Analysis	Dept Co-Sales Correlation	Top 15 association pairs for bundling/layout
Demand Forecasting	XGBoost + RF + Baseline	XGBoost achieves best RMSE; lag features most predictive

13.2 Most Valuable Insights

1. Seasonality is the dominant sales driver - week-of-year and lag features outperform all economic indicators
2. Holiday impact is real but concentrated - the 7% overall lift obscures much larger department-level spikes
3. Markdowns work but non-linearly - moderate promotional intensity generates lift; deep discounts show diminishing returns
4. Store format strongly mediates performance - one-size-fits-all strategies are suboptimal across A/B/C formats

13.3 Future Improvements

Short-Term:

- Add Prophet model for long-horizon (6-12 month) forecasting with built-in trend changepoints
- Enrich external data with regional competitor pricing and demographic data

Medium-Term:

- Obtain item-level transaction data for true Apriori/FP-Growth market basket analysis
- Build a store-level anomaly monitoring dashboard for real-time flagging

Long-Term:

- Deploy XGBoost model as a REST API (FastAPI + Docker) with automated weekly retraining
- Build an LSTM or Transformer model for sequence-aware department-level forecasting
- Integrate real-time POS data stream for daily demand updates